

NIKHIL SACHAN

Software Engineer | GenAI & LLM Applications

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Professional Summary

Software engineer with 2+ years building distributed backend systems, now extending into GenAI application development. Hands-on experience building LLM-powered applications with LangChain, including a tool-using AI agent, a document Q&A system using Retrieval-Augmented Generation (RAG), and a resume-to-cover-letter generator — using ChromaDB, Hugging Face embeddings, and the Groq API. Strong foundation in object-oriented design, distributed systems, relational and NoSQL databases, and Agile development practices.

Core Skills

GenAI & LLM: LangChain, AI agents & tool use, Prompt Engineering, RAG pipelines, Vector Databases (ChromaDB), Hugging Face embeddings, Groq API, Gradio

Programming: Python, Java, Node.js, Object-Oriented Design, Data Structures & Algorithms

Systems & Design: Distributed (multi-tiered) systems, Microservices, REST APIs, HLD/LLD

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB (vector), SQL query optimization

Cloud & DevOps: Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Grafana

Practices & Tooling: Agile/Scrum, Kanban, automated testing, Git, GitHub, Kafka, Redis

GenAI Projects

Tool-Using AI Agent (Gradio) —

- Built an AI agent capable of invoking external tools to complete multi-step tasks, with a Gradio interface for interactive use, similar in spirit to "AI Arena"-style agent demos.
- Designed the agent's tool-calling logic to select and invoke the correct tool based on user intent, handling responses and returning results conversationally.

Tech Stack: Python, LangChain, Gradio, LLM APIs, Tool/Function Calling

Document Q&A App (RAG) —

- Built a retrieval-augmented Q&A application that lets users upload a document and ask natural-language questions against its content.
- Implemented a RAG pipeline: document chunking, embedding generation, vector storage/retrieval, and LLM-based answer synthesis grounded in retrieved context.

Tech Stack: Python, LangChain, RAG, Vector Database, Embeddings

Resume-to-Cover-Letter Generator —

- Built an app that takes a resume and job description as input and generates a tailored cover letter using an LLM.
- Used ChromaDB for vector storage, Hugging Face embeddings for semantic matching between resume and job description, and the Groq API for fast, free LLM inference.

Tech Stack: Python, LangChain, ChromaDB, Hugging Face Embeddings, Groq API, RAG

Experience

Software Engineer — OnlineSportstech Apr 2024 – Present

- Designed and developed backend services (HLD/LLD) for real-time sports scoring, live commentary, and meeting platforms using Spring Boot, REST APIs, WebSockets, and Kafka.
- Built and maintained distributed, multi-tiered architecture with PostgreSQL, Redis caching, and Kafka-based event pipelines to support high-frequency real-time updates.
- Completed code reviews and automated testing to maintain code quality standards, collaborating with peers on design and implementation decisions.
- Managed CI/CD pipelines and containerized deployments (Docker, Kubernetes) on Azure, with Prometheus/Grafana for monitoring application performance and reliability.

- Worked in an Agile/Scrum environment, participating in sprint planning, iterations, and user-story-driven development.

Tech Stack: Java, Spring Boot, PostgreSQL, Kafka, Redis, Azure, Docker, Kubernetes, GitHub Actions

Mobile App Developer — Javin Enterprises *Jan 2023 – Nov 2023*

- Delivered 3 high-traffic mobile applications with object-oriented design practices, consistently maintaining a 4.5-star user rating.
- Owned end-to-end release lifecycle including development, testing, and Play Store/App Store deployment.

Tech Stack: Node.js, Express, Flutter, REST APIs, Firebase, MySQL, Git, GitHub

Other Projects

ScoreWaves — Live Sports Scores & News —

- Designed scalable backend architecture using Redis caching, PostgreSQL, and asynchronous processing, reducing API response times by 60%.
- Implemented observability (Prometheus, Grafana, Loki) and supported 50k+ app downloads through Firebase integrations and analytics.

Tech Stack: Spring Boot, Azure, PostgreSQL, Kafka, Redis, Docker, Kubernetes

Education

I.K. Gujral Punjab Technical University (PTU) — B.Tech, Computer Science | 2023